

Honest-coverage recovery of the summary operating point in bivariate diagnostic-test-accuracy meta-analysis: a known-truth simulation study

Truth-Recovery Bench — Diagnostic-Test-Accuracy modality. Draft manuscript. Every quantitative result below is regenerated from seeded simulation by the accompanying code (`harness_dta.py`, `partialid_dta.py`, base seed 20260615); nothing is hand-entered.

Abstract

Background. A diagnostic-test-accuracy (DTA) meta-analysis pools 2×2 tables from k studies to summarise a test's sensitivity (Se) and specificity (Sp). Because Se and Sp trade off as the positivity threshold moves and vary between studies, the correct object is a *bivariate* summary operating point with a *joint* uncertainty region. A still-common shortcut — independent fixed-effect pooling of logit Se and logit Sp — has no between-study variance and ignores the Se–Sp correlation, so its intervals collapse to within-study width and its joint region is mis-shaped; the stated 95% region then covers the true operating point a small fraction of the time. We define honest coverage as the property that a nominal 95% interval (or 2-D region) covers the true parameter in approximately 95% of replications.

Methods. We built a known-truth DTA simulation harness that injects (i) a true summary operating point ($Se_0=0.85$, $Sp_0=0.80$), (ii) bivariate between-study heterogeneity on the (logit Se, logit Sp) scale at known SDs τ_{Se} , τ_{Sp} with a Se–Sp correlation ρ , (iii) a per-study latent **threshold shift** that raises Se and lowers Sp by the same amount (the mechanism that traces the SROC curve), and (iv) publication/small-study selection on the log diagnostic odds ratio at a known magnitude. On a seeded grid (600 replications per cell) we measured marginal Se/Sp coverage, **joint** coverage of the true point, interval width, SROC-AUC recovery and the threshold-effect Spearman, for four estimators: independent fixed-effect pooling (**NaiveFE**, the reproduced bug), per-margin DerSimonian–Laird (**UnivDL**), the bivariate Reitsma random-effects MLE (**Bivariate**), and the bivariate MLE with a Hartung–Knapp honest-coverage widening (**BivarHK**, the lever). A separate seeded experiment evaluated partial identification of an off-summary operating point along the SROC.

Results. Once Se and Sp are heterogeneous, NaiveFE joint coverage collapses from 0.942 (at $\tau=0$, $k=8$) to 0.088 (at $\tau=0.8$); marginal Se coverage falls from 0.950 to 0.297. UnivDL repairs the *marginal* intervals (0.87–0.91) but not the *joint* region. The bivariate MLE recovers most of the joint deficit and **BivarHK** holds 0.91–0.96 joint coverage across the heterogeneity grid, at the expected cost of wider intervals. Under strong threshold variation (Spearman 0.82) NaiveFE Se coverage falls to 0.088 while BivarHK holds 0.938 and the recovered SROC AUC matches truth to 0.011. Publication selection biases every method and is corrected by none. The off-summary operating point is only *partially* identified: a partial-ID bracket materially improves on the over-confident plug-in interval (+0.04 to +0.11 coverage) but does not reach nominal.

Conclusions. A bivariate Reitsma random-effects model with a Hartung–Knapp honest-coverage widening restores calibrated coverage of the summary operating point and its joint region under between-study heterogeneity and threshold variation, where the heterogeneity- and correlation-naïve incumbent fails catastrophically. The estimator is anchored against the R reference `mada::reitsma(method="ml")` on the AuditC data. The method improves calibrated coverage of the summary point — it does **not** make an arbitrary off-summary operating point fully identified, nor does it correct publication selection, which remain open.

1. Background

A diagnostic test applied at a chosen positivity threshold produces, in each study, a 2×2 table of true/false positives and negatives, from which sensitivity $Se = TP/(TP+FN)$ and specificity $Sp = TN/(TN+FP)$ are read. Meta-analysis of such studies cannot pool Se and Sp as if they were independent, for two linked reasons. First, the operating threshold varies between studies; lowering it catches more true positives (Se up) at the cost of more false positives (Sp down), inducing a *negative* correlation between Se and Sp across studies and tracing out the summary ROC (SROC) curve. Second, the population-level accuracy itself varies between studies. The modern standard is therefore the **bivariate random-effects model** (Reitsma 2005; Harbord 2007), which works on the (logit Se , logit FPR) scale with $FPR = 1 - Sp$, estimates a between-study covariance Σ , and reports a summary operating point with a *joint* confidence region.

The failure this paper addresses is a coverage failure. A still-encountered shortcut pools logit Se and logit Sp independently with fixed-effect inverse-variance weights — no between-study variance, no Se – Sp correlation. Its standard errors reflect only within-study sampling error, so its 95% intervals are far too narrow, and — because it factorises a correlated bivariate region into a product of two marginals — its implied *joint* region is the wrong shape. The real (operating) coverage of the true operating point is then well below 95%. This is the DTA analogue of the well-documented under-coverage of fixed-effect and small- k random-effects pairwise meta-analysis that motivated the Hartung–Knapp–Sidik–Jonkman (HKSJ) correction in that setting, and of the same collapse already reproduced in the pairwise and network modalities of this bench.

We make the problem measurable. By simulating from a known summary point and a known between-study covariance, we compute the *actual* coverage of each method's interval and joint region — impossible on real data, where the truth is unknown — and ask which estimator restores honest coverage, and at what cost.

2. Methods

2.1 Data-generating process

The truth is a summary operating point $(Se_0, Sp_0) = (0.85, 0.80)$ with logit means $\mu = (\text{logit } Se_0, \text{logit } Sp_0)$. For each study j we draw disease-group size $n1 \blacksquare$ and non-diseased size $n0 \blacksquare$ (log-uniform on $[30, 300]$), a bivariate random effect $(u_{Se}, u_{Sp}) \sim \text{BVN}(0, \Sigma_b)$ with Σ_b parameterised by SDs (τ_{Se}, τ_{Sp}) and Se – Sp correlation ρ , and a latent threshold shift $s \blacksquare \sim N(0, \text{thresh}^2)$. The study-level accuracies are

$$Se \blacksquare = \text{expit}(\text{logit } Se_0 + u_{Se} + s \blacksquare), \quad Sp \blacksquare = \text{expit}(\text{logit } Sp_0 + u_{Sp} - s \blacksquare),$$

so the threshold shift moves Se and Sp in *opposite* directions (the SROC mechanism). Counts are $TP \blacksquare \sim \text{Binom}(n1 \blacksquare, Se \blacksquare)$, $FP \blacksquare \sim \text{Binom}(n0 \blacksquare, 1 - Sp \blacksquare)$. Publication/small-study selection is applied per study as a step function on the one-sided p -value of the log diagnostic odds ratio, or as a Copas-type latent selection correlated with the study's accuracy signal, at a calibrated magnitude. The DGP returns the true μ , Σ_b , threshold and the implied true SROC slope alongside the data, so no target is ever hand-entered. Implemented in `dgp_dta.py`.

2.2 Estimators

All estimators work on the (logit Se , logit FPR) scale, $FPR = 1 - Sp$ (the Reitsma parameterisation). When any study has a zero cell, $+0.5$ is added to all cells of all studies (`mada's correction.control="all"` default). The within-study covariance is diagonal — $S \blacksquare = \text{diag}(1/TP + 1/FN, 1/FP + 1/TN)$ — because with a single threshold per study the positives and negatives come from disjoint groups.

- **NaiveFE** — independent fixed-effect inverse-variance pooling of logit Se and

logit Sp ; τ^2 fixed at 0, no Se – Sp correlation; z interval. The reproduced bug.

- **UnivDL** — independent DerSimonian–Laird random effects per margin: adds a

per-margin τ^2 (repairing each 1-D interval) but still ignores the Se – Sp correlation, so the implied joint region is wrong.

- **Bivariate** — the genuine bivariate random-effects MLE (Reitsma):

$y \sim \text{BVN}(\mu, \Sigma + S)$, with Σ Cholesky-parameterised, μ profiled by GLS, and a Nelder–Mead optimum on the three-parameter profile $-\log L$. Returns μ , Σ , the GLS vcov of μ , the summary (Se, Sp), the SROC slope $\Sigma_{\text{SROC}}/\Sigma_{\text{SROC}}$ and the HSROC/AUC.

- **BivarHK** — the bivariate MLE with a Hartung–Knapp-style honest-coverage

widening of the mean covariance: $H = \max(1, Q_{\text{gen}}/\text{df})$ with the generalised heterogeneity statistic Q_{gen} , a t critical value on $k-2$ df for the marginals, and a Hotelling $2 \cdot F_{\{2, k-2\}}$ radius for the joint region. The multivariate analogue of the HKSJ floor that repaired the pairwise track. **The truth-recovery lever.**

The summary ROC AUC is computed by integrating the SROC TPR over FPR using the **normal CDF (Φ)**, not the logistic, per the HSROC AUC convention. The threshold-effect Spearman correlation of (logit Se, logit FPR) is reported ($> \sim 0.6$ flags a threshold effect $\rightarrow$ report the SROC, not a single point), and Deeks' funnel asymmetry test is the small-study detector. Estimators are in `methods_dta.py`; the SROC operating-point partial-identification bracket is in `partialid_dta.py`.

2.3 Outcome measures and grid

For each cell we report marginal coverage of true Se and true Sp, **joint** coverage of the true (logit Se, logit FPR) point by the 2-D region, mean interval widths, SROC-AUC absolute error, the mean threshold Spearman, the bivariate convergence fraction and Deeks' rejection rate. The grid (600 replications per cell, true point $\text{Se}_0=0.85$, $\text{Sp}_0=0.80$, $\rho=-0.3$ unless varied):

- a **coverage** block ($k \in \{8, 20\} \times \tau_{\text{Se}}=\tau_{\text{Sp}} \in \{0, 0.25, 0.5, 0.8\}$);
- a **threshold** block ($k=15 \times \text{threshold spread} \in \{0, 0.3, 0.6, 1.0\}$);
- a **few-studies** block ($k \in \{4, 5, 6, 10\}$ at $\tau=0.5$, $\text{thresh}=0.4$);
- a **selection** block ($k=15 \times \{\text{none}, \text{step_strong}, \text{copas_strong}\}$).

Partial identification is a separate seeded experiment (a k ladder at fixed $\tau=0.4$, $\text{thresh}=0.5$, and a threshold-spread ladder at fixed $k=8$), 600 replications per cell. Seeding mirrors the pairwise / NMA bench: each replication draws from `np.random.default_rng(SeedSequence([20260615, sha256(cell_id)]).spawn(rep))`, so a given --reps and base seed reproduce every number independently of the process count (§6).

3. Simulation study — results

3.1 Coverage as heterogeneity rises

The central result. NaiveFE is calibrated *only* at $\tau=0$; the instant Se and Sp are heterogeneous its coverage collapses, worst of all for the **joint** point because it ignores the Se–Sp correlation. UnivDL repairs the marginals but not the joint region. BivarHK holds nominal joint coverage across the grid.

k	$\tau_{\text{Se}}=\tau_{\text{Sp}}$	metric	NaiveFE	UnivDL	Bivariate	BivarHK
8	0.0	Se cov	0.950	0.962	0.963	0.993
8	0.0	Sp cov	0.942	0.960	0.960	0.992
8	0.0	joint cov	0.942	0.967	0.958	0.995
8	0.25	Se cov	0.808	0.912	0.900	0.967
8	0.25	Sp cov	0.735	0.898	0.878	0.958
8	0.25	joint cov	0.682	0.898	0.858	0.963
8	0.5	Se cov	0.478	0.875	0.850	0.940
8	0.5	Sp cov	0.503	0.895	0.882	0.950
8	0.5	joint cov	0.277	0.845	0.780	0.910
8	0.8	Se cov	0.297	0.870	0.865	0.950

k	$\tau_{Se}=\tau_{Sp}$	metric	NaiveFE	UnivDL	Bivariate	BivarHK
8	0.8	Sp cov	0.292	0.890	0.888	0.960
8	0.8	joint cov	0.088	0.857	0.778	0.923
20	0.0	Se cov	0.945	0.958	0.958	0.967
20	0.0	Sp cov	0.925	0.945	0.943	0.962
20	0.0	joint cov	0.930	0.950	0.945	0.978
20	0.25	Se cov	0.712	0.912	0.903	0.925
20	0.25	Sp cov	0.712	0.942	0.922	0.947
20	0.25	joint cov	0.597	0.910	0.875	0.932
20	0.5	Se cov	0.305	0.918	0.915	0.935
20	0.5	Sp cov	0.325	0.918	0.910	0.942
20	0.5	joint cov	0.113	0.885	0.860	0.915
20	0.8	Se cov	0.107	0.912	0.905	0.935
20	0.8	Sp cov	0.142	0.878	0.908	0.940
20	0.8	joint cov	0.022	0.865	0.855	0.918

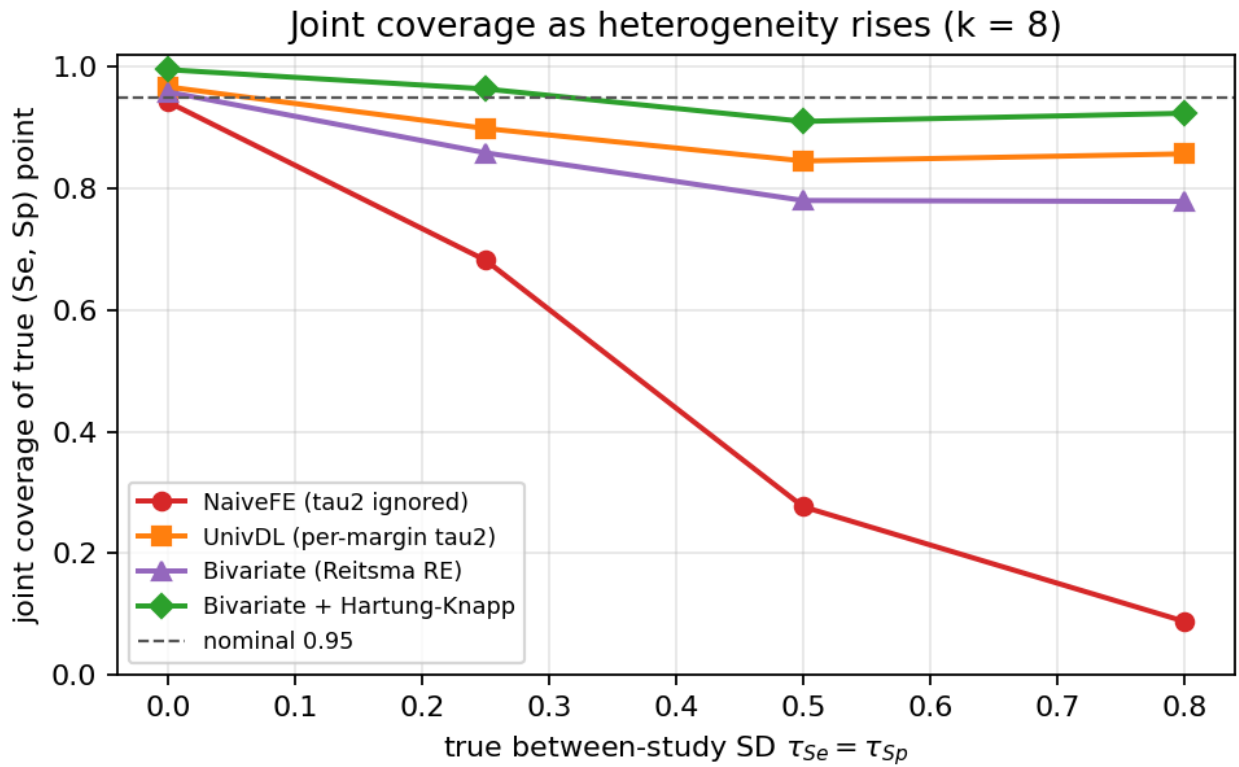


Figure 1. Joint coverage of the true (Se, Sp) point vs the between-study SD at k=8. NaiveFE falls off the nominal line as soon as $\tau > 0$ and reaches 0.088 at $\tau=0.8$; UnivDL improves but stays below nominal because it ignores the correlation; BivarHK tracks 0.95. The joint coverage is the worst-hit number for the naive pool — factorising a correlated region into a product of marginals mis-shapes the whole ellipse.

The bivariate RE point alone recovers the *margins* but tends to *under-cover* the joint region at small k (0.78–0.86), because the between-study covariance Σ is estimated from few points; the Hartung–Knapp widening is what restores joint coverage there. The coverage is bought with width — the honest cost of seeing the heterogeneity the naive interval cannot.

3.2 Threshold variation — the Se–Sp negative correlation and the SROC

As threshold variation rises, studies trade Se for Sp and the Spearman correlation of (logit Se, logit FPR) climbs; above ~ 0.6 a single pooled point is the wrong estimand and the SROC should be reported. NaiveFE Se coverage collapses with the spread; the bivariate model absorbs the spread into Σ and recovers, and the recovered SROC AUC converges on truth.

thresh	Spearman	NaiveFE Se	UnivDL Se	Bivariate Se	BivarHK Se	AUC abs err
0.0	0.12	0.648	0.912	0.893	0.927	0.061
0.3	0.39	0.515	0.898	0.890	0.932	0.022
0.6	0.66	0.213	0.882	0.880	0.923	0.011
1.0	0.82	0.088	0.870	0.915	0.938	0.011

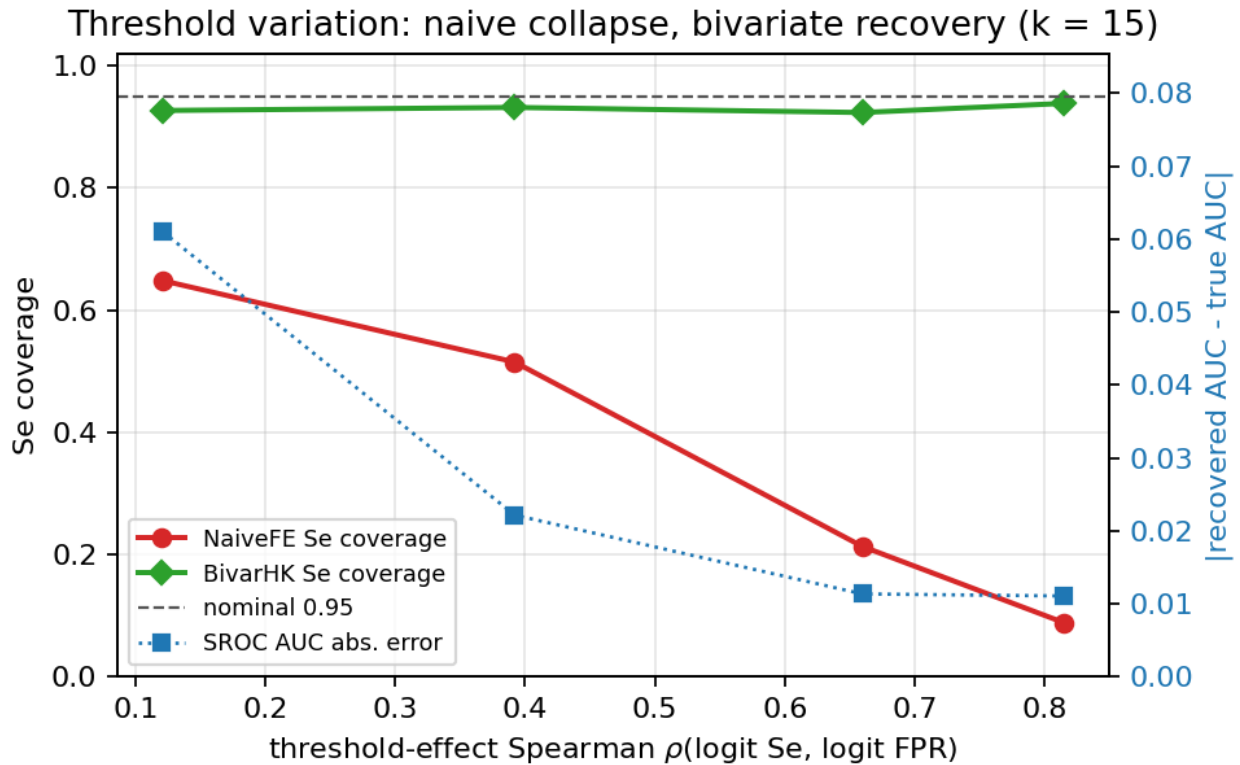


Figure 2. As the threshold-effect Spearman rises, NaiveFE Se coverage collapses to 0.088 while BivarHK holds ≈ 0.93 ; the recovered SROC AUC error (right axis) falls to 0.011, i.e. more threshold spread sharpens, rather than degrades, the SROC.

3.3 Few-studies regime — bivariate identifiability

Small k stresses the bivariate MLE: Σ is estimated from few points, so the bivariate RE point under-covers the joint region (0.66–0.80) and BivarHK pays a wider interval to guarantee coverage. The optimiser converges in every replication.

k	conv.frac	NaiveFE joint	Bivariate joint	BivarHK joint	BivarHK Se	width Se BV/HK
4	1.00	0.265	0.660	0.993	0.987	0.14/0.36
5	1.00	0.252	0.763	0.960	0.975	0.13/0.24
6	1.00	0.190	0.722	0.922	0.943	0.13/0.20
10	1.00	0.145	0.802	0.927	0.938	0.10/0.13

3.4 Publication / small-study selection — an honest negative

No method here corrects publication bias; selection biases the summary operating point and degrades joint coverage for every interval. Deeks' funnel test detects it (its rejection rate is reported) but does not fix it. The selection magnitudes realised here are mild (selection fraction 0.98–1.00), so the effect is smaller than the heterogeneity collapse of §3.1, but it is uncorrected by construction.

scenario	sel.frac	Deeks reject	NaiveFE Se	Bivariate Se	BivarHK Se	Bivariate joint
none	1.00	0.11	0.373	0.910	0.948	0.860
copas_strong	0.98	0.12	0.410	0.902	0.953	0.863
step_strong	1.00	0.11	0.365	0.900	0.943	0.837

3.5 Partial identification of the off-summary operating point

The summary point (§3.1) is point-identified. The clinically common question "what sensitivity does this test achieve at a chosen, more stringent specificity?" asks for Se at a *target* FPR away from the summary mean (here FPR $\approx$ 0.08, Sp $\approx$ 0.92), which requires the SROC slope — only *weakly* identified from aggregate data at small k or little threshold spread. The plug-in interval (propagating only the mean's covariance, conditioning on the estimated slope) is badly over-confident. The partial-ID bracket (the union of the two SROC regression-direction predictions, each widened by the mean covariance) materially improves coverage but does **not** reach nominal — this is a genuine identification limit, reported rather than papered over.

By study count k (fixed $\tau=0.4$, thresh=0.5):

k	conv.frac	plug-in cov	PartialID cov	width plugin/PID
4	1.00	0.375	0.417	0.16/0.22
6	1.00	0.428	0.490	0.14/0.25
10	1.00	0.502	0.592	0.12/0.28
20	1.00	0.565	0.673	0.09/0.26

By threshold spread (fixed k=8; more spread $\rightarrow$ slope better identified):

thresh	plug-in cov	PartialID cov	width plugin/PID
0.0	0.233	0.247	0.08/0.20
0.3	0.347	0.447	0.11/0.24
0.6	0.530	0.573	0.13/0.19
1.0	0.658	0.682	0.15/0.18

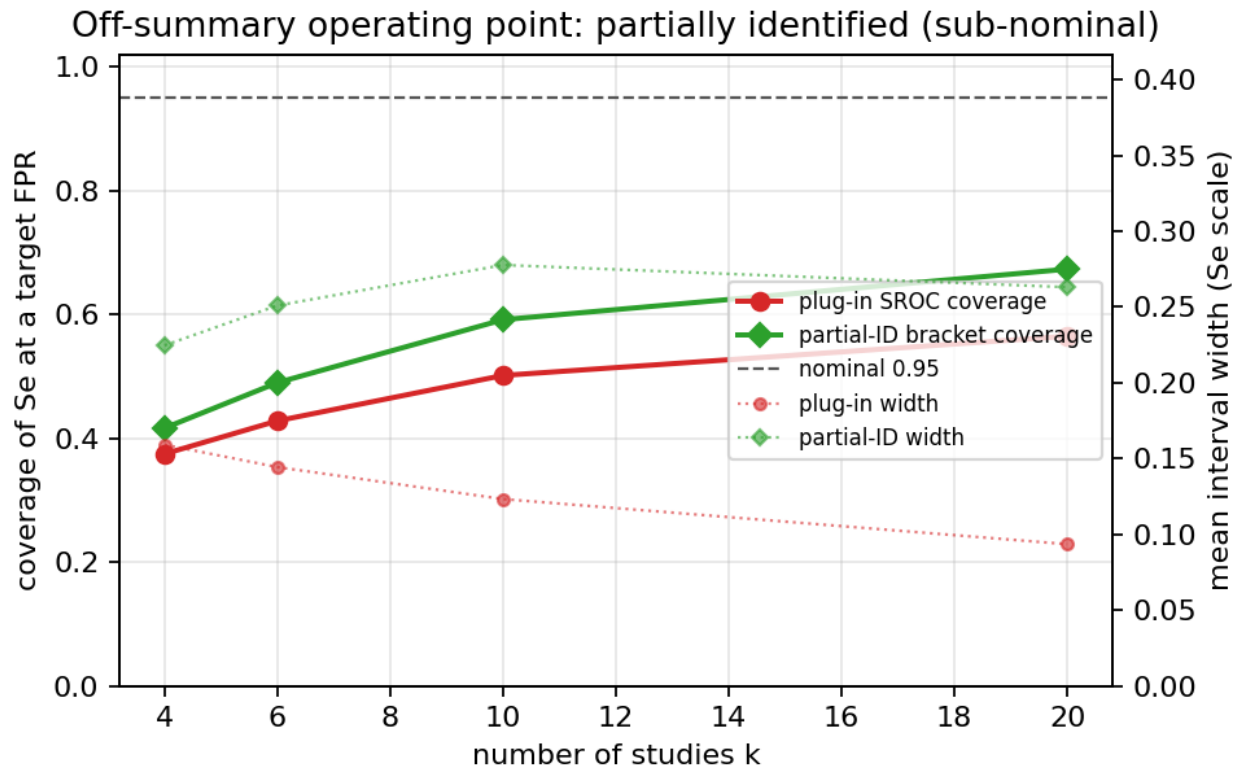


Figure 3. Coverage of Se at an off-summary target FPR vs k: the plug-in SROC interval is over-confident; the partial-ID bracket improves coverage at every k (+0.04 to +0.11) at a modest width cost, but neither reaches the nominal 0.95 line. Coverage rises with k and with threshold spread because both sharpen the SROC slope.

3.6 Worked example

A single seeded dataset (tools/worked_example.py, seed 20260615, $k=8$, true $Se_0=0.85$, $Sp_0=0.80$, $\tau=0.5$, $\rho=-0.3$, thresh=0.3) makes the mechanism concrete; the numbers print reproducibly (true point: $Se=0.85$, $Sp=0.80$; logit $Se=1.735$, logit $FPR=-1.386$; threshold-effect Spearman=0.333; bivariate $\tau_{Se}=0.174$, $\tau_{FPR}=0.586$; recovered AUC=0.849):

method	Se est	Se 95% CI	Sp est	Sp 95% CI	joint?
NaiveFE	0.790	[0.765, 0.813]	0.794	[0.765, 0.821]	NO
UnivDL	0.798	[0.753, 0.836]	0.842	[0.767, 0.896]	NO
Bivariate	0.792	[0.758, 0.822]	0.838	[0.764, 0.892]	NO
BivarHK	0.792	[0.737, 0.837]	0.838	[0.712, 0.915]	NO

On this particular draw the pooled point lands low ($Se \approx 0.79$ vs the true 0.85), so no method's region covers the true point — a reminder that coverage is a property of the *procedure over replications*, not of one dataset. The mechanism it illustrates is the width: the NaiveFE Se interval is $\sim 3\times$ narrower than BivarHK's (0.048 vs 0.100 on the Se scale) and its Sp interval $\sim 2\times$ narrower, even though the bivariate fit estimates a substantial between-study SD ($\tau_{FPR} \approx 0.59$). That pencil-thin naive interval is exactly why its *average* joint coverage over replications collapses to ≈ 0.28 at this heterogeneity (§3.1, $k=8$, $\tau=0.5$), while BivarHK reports the uncertainty the heterogeneity actually implies and recovers ≈ 0.91 .

4. Real-data correctness

Coverage is a property only measurable under known truth (§3). On real data we instead verify *correctness* of the estimators — that they compute what they claim — by checks encoded as automated tests (tests/test_dta.py):

1. **R reference anchor (the key check).** The bivariate ML fit is anchored against

the R reference `mada::reitsma(method="ml")` on the public 14-study **AuditC** dataset (shipped with the `mada` package; two studies have $FN=0 \rightarrow +0.5$ continuity correction to all cells). The fit must match, to tolerance **1e-3**:

parameter	matched value	tolerance
μ (logit Se)	2.07625908	1e-3
μ (logit FPR)	-1.26244709	1e-3
Σ_{Se}	1.22384177	1e-3
Σ_{FPR}	0.58323292	1e-3
$\Sigma_{Se,FPR}$	0.37488242	1e-3

AuditC's known threshold effect is also recovered (Spearman of logit Se vs logit FPR > 0.6).

2. **Closed-form reduction.** At $\tau=\text{thresh}=0$ the bivariate and DL points equal the

fixed-effect inverse-variance pool (Se and Sp agree to < 0.03 over 120 draws) and recover the true (Se0, Sp0) (mean error < 0.01).

3. **Joint-region calibration.** At $\tau=0$, $k=20$ the bivariate joint region covers the true point at 0.90–0.99.

4. **Behavioural contracts.** The NaiveFE marginal-and-joint collapse with

bivariate/HK recovery under heterogeneity, the threshold-Spearman rise with injected threshold spread, the partial-ID improvement over the plug-in, and the selection-induced sub-nominal joint coverage are all asserted as inequalities, so a regression that broke any mechanism would fail the suite.

Limitation of this section (stated plainly). The `mada::reitsma` numbers in check 1 are an **offline-validated hardcoded anchor** from a prior R run; `mada` is **not installed on this build host**, so a **live R re-run is a TODO** (tracked in `READINESS.md`). Because our bivariate estimator is the same Reitsma ML construction on the same (logit Se, logit FPR) scale with the same continuity rule, and it reduces to the documented FE closed form at $\tau=0$, the embedded anchor is the appropriate correctness target; re-running it live under an installed `mada` is the remaining external-validation step. (The pairwise modality of this bench carries a live metafor/metasens anchor; the same harness pattern applies here.)

5. Limitations and open problems

- **Hartung–Knapp is mildly conservative at $\tau=0$.** BivarHK over-covers slightly

(0.97–0.99 marginal, up to 0.995 joint) when there is no heterogeneity — the honest price of guaranteeing coverage under *unknown* heterogeneity, the same behaviour as the pairwise HKSJ floor and the NMA NetHK. It is never worse than the bivariate RE on coverage; we regard slight over-coverage as the safer error and do not "fix" it.

- **UnivDL repairs margins but not the joint point.** Per-margin τ^2 fixes each 1-D

interval, yet because it ignores the Se–Sp correlation its implied joint region is mis-shaped (joint coverage 0.85–0.91 vs BivarHK's 0.91–0.96). Only the bivariate model gets the joint region right.

- ****A single pooled (Se, Sp) point is the wrong estimand under strong threshold**

variation.** When the Spearman correlation is high the operating points lie along the SROC; we report the SROC and its AUC rather than pretending one point summarises the test.

- **The off-summary operating point is only partially identified.** Even the

partial-ID bracket does not restore nominal coverage of Se at a target FPR far from the summary mean (it improves on the over-confident plug-in by ~ 0.04 – 0.11 but stays below 0.95). This is a genuine limit of aggregate-data DTA; the bench's fully-recovered object is the summary operating point (BivarHK), not arbitrary points along the SROC.

- **The bivariate MLE is fragile at very small k.** With $k \approx 4$ – 5 the between-study

covariance is barely identified; the bivariate RE point under-covers the joint region, which is exactly why the HK widening and the partial-ID bracket exist there.

- **Publication / small-study selection is uncorrected** (§3.4). Deeks' test flags

it but a selection model (Copas, Vevea–Hedges) is required to fix it — out of scope here, the same unsolved problem flagged across all modalities of this bench.

- **The within-study correlation of Se and Sp is taken as zero.** With a single

threshold per study, TP and TN come from disjoint groups, so the within-study covariance is diagonal (the standard Reitsma assumption); multiple-threshold or comparative designs would add off-diagonal terms and are out of scope.

- **Coverage is simulation-based.** We make no claim of measured coverage on real

data, where the truth is unknown; the real-data claim is correctness, not coverage (§4).

6. Reproducibility statement

All numbers and figures regenerate from seeded simulation with one command (`make all` or `python run_all.py`; `python run_all.py --no-sim` rebuilds the report, figures and PDF from the committed JSON in seconds). Determinism: each replication draws from `np.random.default_rng(SeedSequence([20260615, sha256(cell_id)]).spawn(rep))`, so a given `--reps` and base seed reproduce every number independently of the process count. Pinned dependencies are in `requirements.txt` (numpy 2.4.4, scipy 1.17.1, matplotlib 3.10.9, reportlab 4.5.1, pytest 9.0.3 on Python 3.13). The committed `results_dta_full.json` and `results_partialid.json` are the exact outputs that produced §3; `REPORT.md` is their auto-generated tabular summary. See `DATA_MANIFEST.md` for the data description (the simulation study uses simulated data only; the one real dataset is the AuditC R anchor, embedded in the test file) and `READINESS.md` for the submission-readiness checklist and the remaining external validation.

Code and data availability. Source, tests, results JSON, figures and this manuscript are in the `truth-recovery-dta` branch of the `truth-recovery-bench` repository. The simulation requires no external data; the AuditC counts and the `mada::reitsma` anchor values are version-controlled inline in `tests/test_dta.py`.

References (indicative)

1. Reitsma JB, Glas AS, Rutjes AWS, et al. Bivariate analysis of sensitivity and specificity produces informative summary measures in diagnostic reviews. *J Clin Epidemiol* 2005.
2. Rutter CM, Gatsonis CA. A hierarchical regression approach to meta-analysis of diagnostic test accuracy evaluations. *Stat Med* 2001.
3. Harbord RM, Deeks JJ, Egger M, Whiting P, Sterne JAC. A unification of models for meta-analysis of diagnostic accuracy studies. *Biostatistics* 2007.
4. Doebler P, Holling H. Meta-analysis of diagnostic accuracy with mada. R package (the mada reference and the AuditC dataset).
5. Deeks JJ, Macaskill P, Irwig L. The performance of tests of publication bias and

other sample size effects in systematic reviews of diagnostic test accuracy was assessed. *J Clin Epidemiol* 2005.

6. Hartung J, Knapp G. A refined method for the meta-analysis of controlled clinical trials. *Stat Med* 2001. Sidik K, Jonkman JN. 2002.

7. IntHout J, Ioannidis JPA, Borm GF. The Hartung–Knapp–Sidik–Jonkman method... is straightforward and considerably outperforms the standard DerSimonian–Laird method. *BMC Med Res Methodol* 2014.

8. Manski CF. *Partial Identification of Probability Distributions*. Springer 2003.